

Fengjie Hong

Third-year Undergraduate Student

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SUMMARY

Third-year undergraduate at the Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University, with a strong background in physics. Currently working on diffusion language models and reinforcement learning.

EDUCATION

Tsinghua University <i>Institute for Interdisciplinary Information Sciences (IIIS)</i> GPA: 3.9/4.0	Beijing, China Sep. 2024 – June 2028 (Expected)
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RESEARCH INTERESTS

Broad interests in machine learning, with a current focus on diffusion models, particularly diffusion language models and reinforcement learning for these models. Open to exploring other areas of machine learning.

RESEARCH EXPERIENCE

Undergraduate Research - Working with Haozhe Jiang (UC Berkeley) on diffusion language models and reinforcement learning. - Participate in Prof. Kaifeng Lyu's group meetings at Tsinghua University.	Present
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TECHNICAL SKILLS

- Programming Languages:** Python, C++
- Frameworks & Tools:** PyTorch, NumPy, Git, LaTeX
- Languages:** Chinese (Native), English

ACADEMIC PROJECTS

LABOTEX <i>Course: Introduction to Large Language Model Applications</i> - In LABOTEX, we present an agentic assistant designed to help researchers generate lab reports in LaTeX format easily and efficiently. - We build 4 agents to read instruction books, write introduction section, process experiment data with tables and plots, and integrate all parts into a complete LaTeX report. These agents interact with each other through the intermediate files they generate. - LABOTEX can generate nice LaTeX reports for General Physics' experiments.	Spring 2025
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WAYF <i>Course: Natural Language Processing</i> - In WAYF (Where Are You From), we present a classifier to detect translationese and fine-tune Qwen3-4B using the classifier to reduce translationese. - For the classifier, we do transfer learning on BERT. We use GRPO to fine-tune Qwen3-4B, with the reward designed from the classifier's output. - Our classifier outperforms zero-shot LLMs on translationese detection.	Autumn 2025
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ME <i>Self-Motivated</i> - To investigate differences between Marx and Engels, we select their writings and analyze semantic differences in key concepts using Word2Vec. - Some differences can roughly be seen.	Autumn 2025
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Realtime Simulation <i>Course: Advanced Computer Graphics</i> - We present a realtime fluid-rigid two way coupling and rendering system. - We use PBD style method for simulation and NVIDIA Warp for multi-threading GPU acceleration. - Achieved realtime movement control.	Autumn 2025
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CVPR <i>Course: Type-safe Modern System Practice</i> - We present Centralized Visual Package Router (CVPR), a typesafe front-end and back-end system for visualized package management. - We build the system with functional programming paradigms and type-safe design principles.	Autumn 2025
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- Emotion Visualization** | *Self-Motivated* 2026
- Built a facial emotion recognition pipeline combining RetinaFace detection with a pretrained ViT classifier, supporting seven emotion categories and per-face attention visualization.
 - Implemented enhanced CLS attention maps and integrated training-free test-time register tokens to reduce attention artifacts, with an interactive web interface and batch inference.
- VideoEditor** | *Diffusion-Based Video Editing* 2026
- Contributed to a video object editing system supporting text prompts, reference images, and sketches through FLUX/AnyDoor first-frame editing and VideoPainter/CogVideoX propagation.
 - The system combines sketch-to-image generation and object-mask tracking with React and Gradio interfaces for interactive editing.

HONORS & AWARDS

- **Tsinghua University Merit Student** 2026
- **Friends of Tsinghua - Weichen Scholarship** 2025
- **Tsinghua University Comprehensive Excellence Scholarship** 2025
- **Freshman Scholarship** 2024
- **Gold Medal in 39th CPhO, 40th CPhO, 24th APhO** 2022-2024